QUALIFICATION STANDARD
AND PLAN (QSP)

Submarine Sonar Watch Leader
NQual (AKYG)

MITE ID
Residential Phase – 123359

Regular Force: Sonar Operator (00324)
Primary Reserve: (N/A)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): Naval Fleet School (Atlantic) (NFS(A))

MARPAC RDIMS #1402031
Chap 1 & 2 approved 13/12/2023
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RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0	Major Update with conversion from 3 Chap. to 4 Chap.	13/12/2023	NTG HQ
1.01	Conversion of TP from 3 Chap. to 4 Chap.	18/01/2024	NFS(A)

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Personnel and Training Group (CNPTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the USQ, Submarine SONAR Watch Leader published 01 Sep 12 A-PD-055-002/PP-002.

3. This QSP publication is effective on receipt and supersedes Submarine SONAR Watch Leader QSP revision 2.1, dated 10 Feb 21. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

1. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened February 2021 at CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened via various working group from February – November 2021, at CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

2. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

3. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500 is required.

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QUALIFICATION STANDARD

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CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Sonar Operator to perform their duties as a Submarine Sonar Watch Leader. Possession of this qualification will allow personnel to supervise sonar operations at sea and alongside.

GRANTING OF NQUAL

2. A trainee will achieve the qualification AKYG Submarine Sonar Watch Leader when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. Nil.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

   a. BSC Qualified (AILS);

   b. Submarine Sonar Op (AILO);

   c. Minimum Rank S1; and

   d. Minimum of 45 sea days onboard Victoria Class Submarine upon AILO completion.

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CHAPTER 2 - PERFORMANCE OBJECTIVES

PO 001 – COMPILE ASW TATICAL PICTURE

1. Performance: Compile ASW Tactical Picture

2. Conditions:

   a. Given:

      (1) Cue: assigned mission and priority at sea;

      (2) Standalone computer (at sea) with references and publications;

      (3) Victoria Class Submarine with sonar set;

      (4) Sonar Watch personnel;

      (5) Supervision (indirect);

      (6) Immediate Command feedback on job performance; and

      (7) Assistance (for clarification and additional personnel if required).

   b. Denied:

      (1) External communication, including internet and intranet (while at sea); and

      (2) Assistance in performance of job duties.

   c. Environment:

      (1) At sea (Victoria Class Submarine):

         (a) Fixed environment (little-no physical movement or adjustment of workspace available); and

         (b) Immediate response required to sonar contacts; and

      (2) Alongside/ashore:

         (a) Office environment (military facility);

         (b) Victoria Class Submarine (alongside); and

         (c) VCSTT for continuation training.

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3. Standard: The Submarine Sonar Watch Leader shall compile ASW tactical picture (E0005), to include:

a. Displaying Intelligence via UWW Brief (B0035) by:

(1) Preparing an Operations Brief for assigned mission by:

(a) Interpreting oceanographic data (F0045);

(b) Determining oceanographic effects on acoustic operations (C0150);

(c) Establishing capabilities and limitations of towed array operations in an operation area (C0160);

(d) Establishing capabilities and limitations of threat contacts;

(e) Calculating acoustic sensor range information (C0115); and

(f) Creating recommendations for mission; and

(2) Presenting an Operations Brief, to include:

(a) Introduction;

(b) Classification level;

(c) References;

(d) Oceanography;

(e) Threat platforms; and

(f) Recommendations;

b. Supervising the operation of the Sound Room Team by:

(1) Conducting a watch turnover as oncoming watch IAW A1 by:

(a) Confirming function check of sensors:

i. 2008;

ii. Echo Sounder; and

iii. 2004;

(b) Reporting to Officer of the Watch (OOW) for Watch Briefing;

(c) Verifying Fire Control Display and sonar picture for alignment;

(d) Verifying functionality of BQQ-10;

(e) Verifying Sonar State Board is current;

(f) Reporting for briefing of sonar picture and pertinent information with offcoming Sonar Watch Leader; and

(g) Assuming Sonar Watch Lead;

(2) Conducting a watch turnover as offcoming watch IAW A1 by:

(a) Briefing oncoming Sonar Watch Leader of the current state, to include:

i. Sensor status of 2008, Echo Sounder, and 2004;

ii. Sonar picture and current contacts; and

iii. Other pertinent information (e.g. recent occurrences and upcoming intentions); and

(b) Reporting watch turnover to OOW to include the following information:

i. Sound Room relieved by;

ii. Confirm all sonar and fire control information is correct; and

iii. Status of equipment turned over;

c. Monitoring operation of all submarine sonar equipment (CT0235) IAW A2;

d. Supervising Sound Room operators during drills and procedures IAW A2, to include:

(1) Correlate active / passive / non-acoustic information (C0105), to include:

(a) Performing Passive Aural Listening (PAL) (CT0195);

(b) Localizing passive acoustic contacts (C0085);

(c) Tracking acoustic data (C0070);

(d) Classifying passive acoustic data (C0080);

(e) Amplifying track information (E0020);

(f) Analyzing passive acoustic data (C0065); and

(g) Evaluating correlated acoustic information (F0005);

(2) Directing towed array system operations (C0155), as delegated, to include:

(a) Performing Passive Aural Listening (PAL) (CT0195);

(b) Localizing passive acoustic contacts (C0085);

(c) Tracking acoustic data (C0070);

(d) Resolving contact ambiguity from towed array;

(e) Classifying passive acoustic data (C0080);

(f) Amplifying track information (E0020); and

(g) Analyzing passive acoustic data (C0065); and

(3) Providing recommendations to Command based on assessment of situation (Situation, Assessment, Recommendation / SAR); and

e. Directing the maintenance of operational logs (B0055) IAW A3, to include:

(1) Recording acoustic intelligence data (F0010);

(2) Sonar Logger;

(3) SRF 203; and

(4) Maintaining sonar stateboard (B0015).

4. References: A1, A2, and A3.

5. Task Numbers: B0015, B0035, B0055, C0065, C0070, C0080, C0085, C0105, C0115, C0150, C0155, C0160, CT0195, CT0235, E0005, E0020, F0005, F0010, and F0045.

6. Remarks: Nil.

TRAINING PLAN

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CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. Residential Training was selected for this course.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 23 training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
MS – PO1	00324	Sub Sonar Watch Leader (AKYG)	1	All

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period
ALDE qualified NWO	1	Act as Command for UWW Brief (EC 001.01)	EC 001.01	8
Lt(N) NWO	1	Act as OOW for EC 001.02	EC 001.02	2
Lt(N) NWO	1	Act as OOW for EC 001.03/PC 001	EC 001.03 / PC 001	26

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. Nil limitations are identified.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

   a. Optimum – 4;

   b. Maximum Load – 6; and

   c. Minimum – 2.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by the Career Manager.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the trainee's strengths and weaknesses with specific comments on any PO for which the trainee demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – COMPILE ASW TACTICAL PICTURE

EO 001.01 DISPLAY INTELLIGENCE VIA UWW BRIEF

1. Performance: Display intelligence via UWW brief.

2. Conditions:

   a. Given:

      (1) Cue: assigned mission and priority (scenario);

      (2) Standalone computer with references and publications;

      (3) Trainer with sonar set;

      (4) Sonar Watch personnel;

      (5) Supervision (indirect);

      (6) Immediate Command feedback on job performance;

      (7) Assistance (for clarification and additional personnel if required);

   b. Denied:

      (1) External communication, including internet and intranet (to simulate being at sea); and

      (2) Assistance in performance of job duties;

   c. Environment:

      (1) Office environment (military facility);

      (2) Victoria Class Submarine (alongside); or

      (3) VCSTT for continuation training.

3. Standard: The trainee shall display intelligence via UWW Brief (B0035) by:

   a. Preparing an Operations Brief (T001) for assigned mission by:

      (1) Interpreting oceanographic data (F0045);

      (2) Determining oceanographic effects on acoustic operations (C0150);

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(3) Establishing capabilities and limitations of towed array operations in an operation area (C0160);

(4) Establishing capabilities and limitations of threat contacts;

(5) Calculating acoustic sensor range information (C0115);

(6) Creating recommendations for mission;

b. Presenting an Operations Brief (T002) including:

(1) Introduction;

(2) Classification level;

(3) References;

(4) Oceanography;

(5) Threat platforms;

(6) Recommendations;

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Explain operations briefing principles and expectations	Method: Interactive Lecture Media: Instructor Environment: Secret Classroom	1 x 50 min	A1, A2
5. Prepare UWW Brief	Method: Practice under supervision Media: Instructor Environment: Secret Classroom	3 x50 min	A1, A2
6. Brief UWW intelligence	Method: Practice Media: Instructor Environment: Secret Classroom	4 x 50 min	A1, A2

5. Total EO Time: Total: 17 x 50 minute periods as follows:

a. 8 x 50 min period; and

b. 9 x 50 min periods – EC (total time).

6. Training and Learning Aids:

   a. Secret Hard Drive with reference material including:
      (1) VSCCO's
      (2) VCSOP's
      (3) CSO's
      (4) NCMP 403
      (5) NCPM 401B
      (6) IETM
      (7) DWAN Trinity Webpage
      (8) CSNI Trinity Webpage
      (9) 4A
      (10) AIG
      (11) Open source internet

   b. Operations Brief template (SECRET);

   c. Scenario / assigned mission;

   d. Guest lecturer to act as Command (minimum ALDE qualified);

7. Remarks: nil.

EO 001.02 CONDUCT A WATCH TURNOVER

1. Performance: Conduct a watch turnover.

2. Conditions:

   a. Given:

      (1) Cue: assigned mission and priority (scenario);

      (2) Standalone computer with references and publications;

      (3) Trainer with sonar set;

      (4) Sonar Watch personnel;

      (5) Supervision (indirect);

      (6) Immediate Command feedback on job performance;

      (7) Assistance (for clarification and additional personnel if required);

   b. Denied:

      (1) External communication, including internet and intranet (to simulate being at sea); and

      (2) Assistance in performance of job duties;

   c. Environment:

      (1) Office environment (military facility);

      (2) Victoria Class Submarine (alongside); or

      (3) VCSTT for continuation training.

3. Standard: The trainee shall conduct a watch turnover to include:

   a. Conducting a watch turnover as oncoming watch IAW A1 by:

      (1) Confirming function check of sensors:

         (a) 2008;

         (b) Echo Sounder; and

         (c) 2004;

      (2) Reporting to Officer of the Watch (OOW) for Watch Briefing;

(3) Verifying Fire Control Display and sonar picture for alignment;

(4) Verifying functionality of BQQ-10;

(5) Verifying Sonar State Board is current;

(6) Reporting for briefing of sonar picture and pertinent information with offcoming Sonar Watch Leader; and

(7) Assuming Sonar Watch Lead;

b. Conducting a watch turnover as offcoming watch IAW A1 by:

(1) Briefing oncoming Sonar Watch Leader of the current state, to include:

(a) Sensor status of 2008, Echo Sounder, and 2004;

(b) Sonar picture and current contacts; and

(c) Other pertinent information (e.g. recent occurrences and upcoming intentions); and

(2) Reporting watch turnover to OOW to include the following information:

(a) Sound Room relieved by;

(b) Confirm all sonar and fire control information is correct; and

(c) Status of equipment turned over;

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Conduct watch turnover	Method: EDI Media: Instructor Environment: Trainer	2 x 50 min	A1, A2

5. Total EO Time: Total: 4 x 50 minute periods as follows:

a. 2 x 50 min period; and

b. 2 x 50 min periods – EC (total time)

6. Training and Learning Aids: minimum ALDE qualified person to act as OOW

7. Remarks:

a. Requires 2x additional staff (1 minimum ALDE qualified); and

b. Trainer needs to be booked in advance and trainer needs to be running with contacts (to be used during handover).

EO 001.03 SUPERVISE SOUND ROOM OPERATORS DURING DRILLS AND PROCEDURES

1. Performance: Supervise sound room operators during drills and procedures.

2. Conditions:

   a. Given:

      (1) Cue: assigned mission and priority (scenario);

      (2) Standalone computer with references and publications;

      (3) Trainer with sonar set;

      (4) Sonar Watch personnel;

      (5) Supervision (indirect);

      (6) Immediate Command feedback on job performance;

      (7) Assistance (for clarification and additional personnel if required);

   b. Denied:

      (1) External communication, including internet and intranet (to simulate being at sea); and

      (2) Assistance in performance of job duties;

   c. Environment:

      (1) Office environment (military facility);

      (2) Victoria Class Submarine (alongside); or

      (3) VCSTT for continuation training.

3. Standard: the trainee shall supervise sound room operators during drills and procedures IAW A2 by:

   a. Correlating active / passive / non-acoustic information (C0105), to include:

      (1) Performing Passive Aural Listening (PAL) (CT0195);

      (2) Localizing passive acoustic contacts (C0085);

      (3) Tracking acoustic data (C0070);

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(4) Classifying passive acoustic data (C0080);

(5) Amplifying track information (E0020);

(6) Analyzing passive acoustic data (C0065); and

(7) Evaluating correlated acoustic information (F0005);

b. Directing towed array system operations (C0155), as delegated, to include:

(1) Performing Passive Aural Listening (PAL) (CT0195);

(2) Localizing passive acoustic contacts (C0085);

(3) Tracking acoustic data (C0070);

(4) Resolving contact ambiguity from towed array;

(5) Classifying passive acoustic data (C0080);

(6) Amplifying track information (E0020); and

(7) Analyzing passive acoustic data (C0065); and

c. Providing recommendations to Command based on assessment of situation (Situation, Assessment, Recommendation / SAR); and

d. Directing the maintenance of operational logs (B0055) IAW A3, to include:

(1) Recording acoustic intelligence data (F0010);

(2) Sonar Logger;

(3) SRF 203; and

(4) Maintaining sonar stateboard (B0015).

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Identify roles and responsibilities of sound room supervision 2. Discuss applicable references and SOPs to include: a) VCSSOs b) VCSOPs c) CSOs d) NCPM 403 e) ITEM f) NCPM 401B g) RSS process h) handling of classified material i) CSNI SOPs	Method: Interactive Lecture Media: Computer Environment: Secret classroom	10 x 50 min	A1, A2, A6, A7, A8, A9
Supervise sound room drills and procedures	Method: Practice under supervision Media: Real Equipment Environment: Trainer	97 x 50 min	A1, A2, A7

5. Total EO Time: Total: 120 x 50 minute periods as follows:

   a. 107 x 50 min period; and

   b. 13 x 50 min periods – EC (total time).

6. Training and Learning Aids: nil.

7. Remarks: ensure the following content is discussed:

   a. Bearing Curves;

   b. Characteristics of Towed Array;

   c. Range Calculations;

   d. Narrow Band Definitions;

   e. Frequency Relationships;

   f. Underwater Telephone;

g. Maintaining Records;

h. Employing Command Information Organization;

i. Close Quarters Drill;

j. Fishing Vessel avoidance; and

k. Water Space Management and Prevention of Mutual Interference.

QUALIFICATION STANDARD ANNEXES

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ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A1	VCSSO AL 6	Victoria Class Submarine Ship's Standing Orders
A2	NCPM 403	Naval Combat Procedures Manual 403
A3	CSRM	Canadian Submarine Records Manual
A4	4A	Active Sonar Classification Guide
A5	AIG	CFCD 113(D) Vol 2 Acoustic Information Guide
A6	NDSOD	National Defense Security Orders and Directives
A7	CSO	Captain's Standing Orders
A8	ITEM	Integrated Electronic Training Manual
A9	NCPM 401B	Naval Combat Procedures Manual 401B Victoria Class Combat Information Organization
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B1	Nil.	
C		**Civilian Publications and Manuals that relate to this Qualification**
C1	Nil.	
D		**Other Publications and Manuals that relate to this Qualification**
D1	Nil.	

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

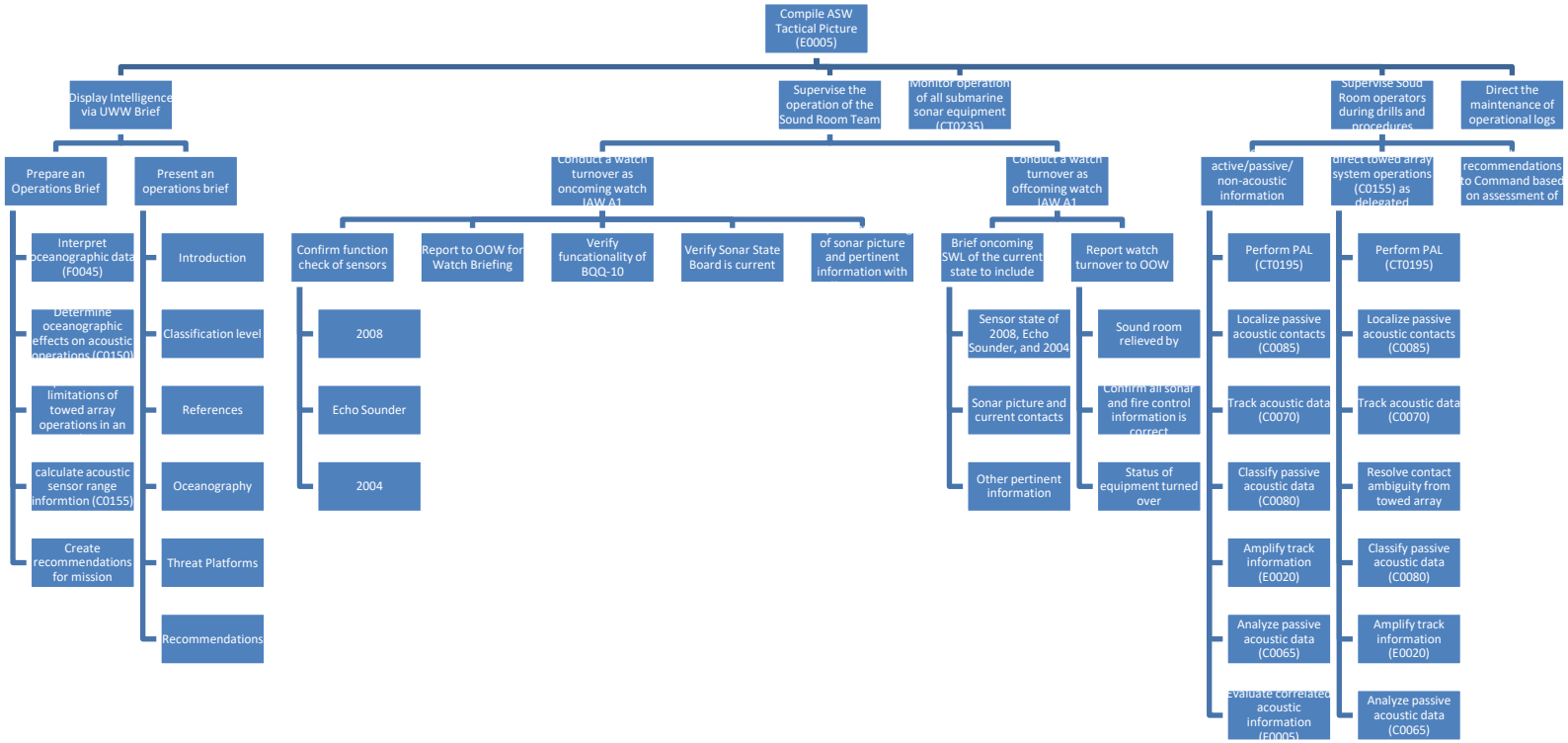
CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
A0005	Perform communications circuit functional checks (comm checks)	NT		Sub-task (Part of daily user checks)
A0010	Operate communications equipment	NT		All members / BSQ
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	NT		All members
A0025	Encode/Decode tactical information (with Authenticate communication transmissions)	NT		Nav Comm
A0030	Authenticate communication transmissions	NT		Nav Comm
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	NT		Above level
B0005	Perform pre-sail checks on equipment	NT		SONAR Tech
B0010	Plot formations, screens and dispositions	NT		Previous qualification
B0015	Maintain stateboards	T	001	
B0020	Enforce Emission control (EMCON) restrictions	NT		

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
B0025	Adhere to Emission control (EMCON) restrictions	NT		Skill
B0030	Contribute to intelligence gathering Ops	NT		Previously trained
B0035	Display intelligence via UWW brief	T	001	
B0040	Disseminate intelligence information	NT		As directed only
B0050	Operate CS	NT		Previously qualified
B0055	Maintain operational logs	T	001	
B0070	Coordinate escape equipment inspection	NT		Secondary duty
B0075	Conduct post inspection procedures	NT		Uncertain task
B0080	Conduct pre-inspection procedures	NT		
B0085	Conduct escape/rescue training	NT		All members
C0020	Operate passive acoustic equipment	NT		Already trained
C0025	Operate ancillary acoustic equipment	NT		Already trained
C0030	Operate oceanographic equipment	NT		Already trained
C0035	Launch underwater expendables	NT		All members
C0040	Analyzing XBT/XSV Forms	NT		Already trained
C0050	Prepare underwater expendables for lauch	NT		All members
C0055	Verify underwater expendables inventories	NT		As directed
C0060	Monitor sonobuoys	NT		Same skill as track contacts
C0065	Analyze passive acoustic data	T	001	Supervise
C0070	Track acoustic contacts	T	001	Supervise
C0075	Annotate recording media	NT		Same as F0010
C0080	Classify passive acoustic data	T	001	Supervise
C0085	Localize passive acoustic contacts	T	001	Supervise
C0100	Construct Contact Area of Probability (AOP)	NT		UC1 level
C0105	Correlate active / passive / non-acoustic information	T	001	
C0115	Calculate acoustic sensor range information	T	001	Supervise
C0120	Calculate array tow speeds, tilt and/or balancing	NT		Not done on the job
C0130	Conduct noise rounds	NT		BSQ
C0135	Perform noise rounds	NT		BSQ

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
C0150	Determine Oceanographic Effects on Acoustic Operations	T	001	Supervise
C0155	Direct towed array system operations	T	001	Supervise
C0160	Establish capabilities and limitations of towed array operations in operation areas (OPAREAS)	T	001	
C0170	Assist with loading/handling of torpedoes	NT		All members
C0195	Perform Passive Aural Listening (PAL)	T	001	Supervise
C0210	Operate vibration analysis equipment	NT		Secondary duty
C0220	Operate One Man Control (OMC)	NT		All members
C0225	Operate Submarine Fire Control System (SFCS)	NT		Different qualification and requirement
C0235	Monitor operation of all submarine sonar equipment	T	001	Supervise
C0240	Supervise Ship Control Console	NT		Not done on job
C0245	Operate electronic fixing aids (EFAs)	NT		Higher level
C0265	Set up the Acoustic Positioning System (APS)	NT		Not done on job
C0280	Operate detection console	NT		Not done on job
C0285	Operate classification console	NT		Not done on job
D0125	Operate Submarine Fire Control System (SFCS)	NT		Different qualification
E0005	Compile ASW tactical picture	T	001	Supervise
E0015	Detect radar contacts	NT		Different trade
E0020	Amplify track information	T	001	Supervise
E0030	Operate Submarine Fire Control System (SFCS)	NT		Different qualification
F0005	Evaluate correlated acoustic information	T	001	Supervise
F0010	Record acoustic intelligence data	NT		AILO course
F0045	Interpret oceanographic data	NT		Previously qualified
G0010	Plan / design training exercises	NT		UC1 level

ANNEX C – QS SCALAR



MARPAC RDIMS #1402031
Chap 1 & 2 approved 13/12/2023
Chap 3 & 4 approved 18/01/2024

ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	**SUBMARINE SONAR WATCH LEADER QSWB ROD**			
Location of Board	**CFB Halifax**			
Dates	**Feb-Nov 2021 (various working groups)**			
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair	Larkin	MD	PO1	NTDC(A)
SME	Shewchuk	SJAM	PO2	NFS(A)
SME	Leveille	SM	MS	HMCS CHICOUTIMI
Advisor	Hart	TD	Lt(N)	NTDC(A)

Item	Details	Action
1	The WG has decided that this QSP should consist of only one PO. Though there would have been the possibility to separate this into two POs, it was decided that all portions of the required performance are so intricately linked that having them under one Performance Objective made the most sense and was most logical for teaching and testing purposes.	
2	The WG has decided that the specific performance covered in EC 001.01 will not be duplicated in the PC as the skills and requirements of turnover brief, including making recommendations to Command, evaluating, interpreting and applying oceanographic data to operational requirements, is amply covered in the performance requirements for Supervise Sound Room Drills and Procedures, the major component of this PC assessment. The WG believes that duplicating this performance in the PC is not required.	
3	The WG compared this course content to the current Sonar Op RQ MS QSP and the overall content and performance requirements align, though this QSP also encompasses the additional expectations and requirements for operating in a Submarine environment that includes additional interactions and expectations for dealing directly with higher Command level.	
4	A note on conditions: though there are conditions outlined in the Performance Objective, the Job Specification for the Basic Submarine Course presents a well-detailed and thorough representation of the conditions and context in which this job is performed and should be considered as part of the holistic conditions for job performance for this qualification as well:	

Item	Details	Action
	“The working environment encountered includes all conditions applicable to the occupation. In addition, employment in submarines provides a set of working conditions for which there is no equivalent in the surface fleet. Submarine duties involve an extremely restrictive environment where noise, extremes of temperature, high humidity, atmospheric contaminants, fluctuating atmospheric pressure and lack of sleep contribute to high levels of physical strain and fatigue. The requirement to live and work for extended periods in close quarters with a large number of other personnel in cramped, unpleasant conditions creates a high degree of emotional stress, which is exacerbated by the fact that a simple error at any time while the submarine is submerged could have catastrophic consequences for the entire crew. Submarine duties may lead to physical disorders arising from extended periods of exposure to atmospheric contaminants such as high partial pressures of carbon dioxide and significant concentrations of carbon monoxide, hydrocarbons, chlorine, stibine, arsine and the combustion products of Freon. Reduction in sight efficiency may result from extended periods of exposure to inadequate illumination and reduction in hearing efficiency may result from exposure to engine and ventilation system noise and rapid changes in air pressure”.	
5	**Previous RODs from V2.0 (7 Nov 2016 approval):** Dates: 14th - 25th November 2011 Location: CFNOS Submarine Division INTRODUCTION	
6	The QSP Board for the Submarine Sonar Control Supervisor was convened during a 2-week period commencing on the 14th November 2011. The purpose of the board was to review the existing SS and QSP of the Submarine Sonar Control Supervisor Course.	
7	The board was convened and given mandate at the direction of: Convening Order - Submarine Senior Sonar Operator Qualification / Standards and Plan (QSP) Review Board AILE 271236ZSEP11 DMTE 222675.	

Item	Details	Action
8	The board complement was as follows: Chairperson — PO1 Martin — CFNOS Stds TDO — Lt Leslie — CFNOS Stds Board Coordinator — CPO2 (RN) Stephenson — CFNOS Sub Div Secretary — LS McDonald — Trinity DMTE Rep — PO2 Pierangeli — DMTE 2-2-4 Standards — PO1 Martin — CFNOS Stds Board Member — PO1 McDonald — HMCS Windsor Board Member — PO1 Snelgrove — CFNOS Board Member — PO2 Murray — ADAC Board Member — MS Drummond — CFNOS Sub Div	
9	Initial discussions included the introduction of all members of the board followed by a brief by the TDO, Lt Leslie, on the process of the board, covering the roles and responsibilities of the board composition, the general aspects, Terms of Reference and a review of CFITES.	
10	14 Nov 11 The SS was reviewed and the following proposals were recommended: a. The board discussed at length that the current AILE QSP is only preparing the Submarine Sonar Control Supervisors to be an adequate Sonar Watch Leader, while the UC1 duties are not clearly defined within the current QSP. This assertion is justified in the current Victoria Class Ship Standing Orders (VCSSO's) where it outlines that the Sonar Watch Leader and the UC1 are two separate jobs while onboard Submarines. b. Therefore the board agreed to create a new specification and QSP entitled Submarine Sonar Watch Leader, with the justification being that the Submarine Sonar Control Supervisor (UC1 equivalent) is no longer a Sound Room Watch Keeper, and that a training gap exists between the Submarine Sonar Op (AILO) Course and the Submarine Sonar Control Supervisor (AILE) Course. c. The board reviewed the previous ROP and the approved SS for the Submarine Sonar Control Supervisor (AILE) Course and created new tasks for the Submarine Sonar Watch Leader Course. After the new SS was created the board decided to proceed with creating performance objectives. Once complete, the board carried on with the instructional analysis phase and completed two performance objectives fully.	

Item	Details	Action
11	REVIEW OF TASKS The Tasks in the approved SS and pervious QSP ROP were reviewed and new tasks were created for the Watch Leader Course: a. T0001 Prepare and Present a Watch Brief b. T0002 Supervise The Operation of the Sound Room Team	
12	REVIEW OF SKILLS No review of skills was required.	
13	REVIEW OF KNOWLEDGE Just like the Tasks, new knowledges had to be created for the new specification: a. K0001 Submarine Oceanography; b. K0002 Command brief requirements; c. K0003 Characteristics and theory of operation of sonar equipment; d. K0004 Voice procedures; e. K0005 Acoustic Analysis; and f. Level 3 for all 5 knowledges.	
14	Train / No-train decisions were made and all tasks were decided to be trained.	
15	The board went through the Scalar Process and established the groupings for the Performance Objectives. The Performance Statements were written and the following PO's were established: a. PO001 - Prepare / Present Operations Brief b. PO002 - Supervise The Operation Of The Sound Room Team	
16	15 November: The board began the instructional analysis phase and completed all enabling objectives. Once complete, the board began work on chapter 4 Lesson Specifications and chapter 3 Assessment of Trainee.	
17	16 November: The board completed work on PO002 including all enabling checks and performance checks and the corresponding checklists were created.	

Item	Details	Action
18	17 November: The board merged PO 003 into PO 002, upon deciding that the two POs should be tested together. The board also discussed the viability of an OJPR after this course, and it was decided that there would not be an OJPR. P01 McDonald expressed concerns about not having an OJPR. In the afternoon, a full review of the QSP was completed as well as researching the references.	
19	18 November: The board convened and amended the Course History Annex and then updated the ROP. A final review was conducted and the board ceased and will continue with the UC1 course for the following week.	

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	
Trg Mgr:	

TRAINING PLAN ANNEXES

MARPAC RDIMS #1402031
Chap 1 & 2 approved 13/12/2023
Chap 3 & 4 approved 18/01/2024

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION	QTY	REMARKS
a.	Victoria Class Submarine Command Team Trainer (VCSCTT)	1	All EOs
b.	Tactical Weapons Simulator (TWS)	1	All EOs
c.	SECRET classroom	1	All EOs
NOTES:			

3. Ammunition (nil)

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons (nil)

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

5. Specialized/Technical Equipment (nil)

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Transport/Vehicles (nil)

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

7. Communication Equipment/Frequencies (nil)

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

8. Training Aids (nil)

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

9. POL Products (nil)

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

10. General Purpose Stores (nil)

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

11. Stationery (nil)

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Personal Equipment (nil)

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: Submarine		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	6	$1,500.0	[redacted]	$9,000	$9,000	$45,000.00
R&Q	6	$72.31	25	$10,846.50	$10,846.50	$54,232.50
Incidentals	6	$17.30	25	$2,595.00	$2,595.00	$12,975.00
Subtotal				**$22,441.50**	**$22,441.50+2400**	**$112,207.50**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	6	$50.00	1	$300.00	$300.00	$1,500.00
Instructional Materials (Printed / Handouts etc.)	6	$30.00	1	$180.00	$180.00	$900.00
Subtotal				**$480.00**	**$480.00**	**$2,400.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
		subtotal		**$22,921.50**	**$22,921.50**	**$114,607.50**
	Total Estimated Costs			**$22,921.50**	**$22,921.50**	**$114,607.50**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	7,700 min (154pd)
ATT Sub-total	**7,700**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	100 min (2pd)
Training Administrative Time (TAT)	100 min (2pd)
Training Travel Time (TTT)	650 min (13pd)
Physical Fitness (PT)	650 min (13pd)
Sub-total	**1500**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Minutes	**9,200 Minutes**
Total Training Days	**23 Days**

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ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
EC001.01	EO 001.01	To confirm the trainee's ability to display intelligence via UWW briefing	Trainees will create and present an UWW brief based on assigned mission	Upon completion of EO 001.01	9 x 50 min periods (total)	- Assigned mission / scenario - Briefing template - ALDE qualified Command	Nil.	8/10 SATs	Remedial Training / Re-test
EC 001.02	EO 001.02	Confirm the trainee's ability to do oncoming and offgoing watch turnovers	Trainee will conduct an oncoming and offgoing watch turnover in the trainer	Upon completion of EO 001.02	2 x 50 min periods (total)	- Trainer (VCSCTT)	Nil.	6/7 SATs	Remedial Training / Re-test
EC 001.03	EO 001.03	Confirm the trainee's ability to supervise sound room drills and procedures at sea	Trainee will supervise sound room drills and procedures	Upon completion of EO 001.03	13 x 50 min periods (total)	- Trainer (VCSCTT or TWS)	Nil.	6/7 SATs	Remedial Training / Re-test

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
PC 001	PO 001	Confirm the trainees ability to supervise the sound room	Trainee will supervise the sound room	Upon completion of EC 001.03	13 x 50 min periods (total)	- Trainer (VCSCTT or TWS)	Nil	8/10 SATs	IAW NTORD 4500-54

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ANNEX I – ASSESSMENT CHECKLISTS

ENABLING CHECK 001.01 DISPLAY INTELLIGENCE VIA UWW BRIEF

EC Time: 420 mins (concurrent) + 30 min debrief

ASSESSMENT ACTIVITY: You have been directed to investigate the oceanographic area and threat contacts for an upcoming mission for the Victoria Class Submarine. The information gathered and analyzed must be presented to Command.

Instructor will provide area of operations and mission objective to trainee.

Trainee will collect information and data from oceanography references, publications, and open source internet. Trainees will compile data into operations brief template on standalone computer with SECRET hard drive.

Instructors will assist trainees as required with data interpretation.

Instructor will direct trainees to present one at a time to “Command” (acted by instructor or NWO as available).

Trainees will individually present the briefing to Command (in group setting). Command/Instructor will receive briefings, ask pertinent confirmation questions and request amplifications on information briefed. Trainee will answer questions as applicable.

Pass/fail criteria: Trainees must achieve 8/10 SATs.

ASSESSMENT RESOURCES:
1) NWO to act as Commander (if available);
2) References and Publications (AIG, 4A);
3) Access to open source internet data files;
4) DWAN access;
5) Operations brief template;
6) Standalone computer; and
7) SECRET hard drive.

IMPACT OF FAILURE: Remedial Training / Re-test

INVIGILATOR NOTES:

Sonar Watch Leader (AKYG) – EC 001.01 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: 8/10 SATs.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	PASS	FAIL	COMMENTS
Brief Components			
Introduction	☐	☐	
Classification	☐	☐	
References	☐	☐	
Oceanography	☐	☐	
Threat Platforms • Detection/counter-detection capabilities • Weapons	☐	☐	
Recommendations	☐	☐	
Briefing relevant to scenario	☐	☐	
Briefing Technique			
Clear, concise, confident	☐	☐	
Answers questions appropriately (if applicable)	☐	☐	
Substantiate recommendations and information as required	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ENABLING CHECK 001.02 – CONDUCT A WATCH TURNOVER

EC Time: 70 mins (concurrent) + 30 min debrief

ASSESSMENT ACTIVITY: You are the Sonar Watch Leader (SWL) onboard a Victoria Class Submarine, the Submarine is operating in Canadian Territorial waters where you expect to detect merchant traffic and various threat platforms, you are about to assume a watch and fulfill your SWL duties.

Part 1: Turnover
The trainee will begin watch turnover (oncoming) by conducting function check of sensors and equipment and then reporting for OOW brief.

Instructor will provide OOW brief (or OOW will be provided by NWO if available) to on-coming trainee.

On-coming Trainee will review Sonar State Board and report for briefing from offcoming SWL (played by instructor if required or trainee off-going assessment).

Out-going trainee will complete watch turnover – offcoming.

Oncoming trainee assumes offcoming SWL position for following trainee.

Pass/fail criteria: 8/10 SATs.

ASSESSMENT RESOURCES:
1) References and publications (AIG, 4A);
2) PC Checklist;
3) Trainer facility including:
    a. Underwater telephone log;
    b. Tactical book;
    c. Chronoscope;
    d. headsets
4) Sonar State Board
5) Trainer Staff (min 1, preferred 2) and support personnel (2 if avail)
   NWO if avail to act as OOW

IMPACT OF FAILURE: Remedial Training / Re-test

INVIGILATOR NOTES:

Sonar Watch Leader (AKYG) – EC 001.01 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: 8/10 SATs.

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2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	PASS	FAIL	COMMENTS
Conduct Watch Turnover IAW VCSSO AL6 - oncoming			
Function check of sensors • 2008 • Echo Sounder • 2004	☐	☐	
OOW Briefing	☐	☐	
Verify Fire Control Display	☐	☐	
Verify functionality of BQQ-10	☐	☐	
Verify Sonar State Board	☐	☐	
Turnover brief from offcoming SWL	☐	☐	
Conduct Watch Turnover IAW VCSSO AL6 - offcoming			
Briefing oncoming SWL on Sensor Status	☐	☐	
Briefing oncoming SWL on Sonar Picture	☐	☐	
Briefing oncoming SWL on other pertinent information as applicable	☐	☐	
Report turnover to OOW • Sound room relieved by • Confirm sonar and fire control information is correct • Status of equipment turned over	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ENABLING CHECK 001.03 – SUPERVISE SOUND ROOM OPERATORS DURING DRILLS AND PROCEDURES

EC Time: 600 mins (concurrent) + 50 min debrief

ASSESSMENT ACTIVITY: You are the Sonar Watch Leader (SWL) onboard a Victoria Class Submarine, the Submarine is operating in Canadian Territorial waters where you expect to detect merchant traffic and various threat platforms, you are about to assume a watch and fulfill your SWL duties.

Pass/fail criteria: 6/7 SATs.

ASSESSMENT RESOURCES:

1) References and publications (AIG, 4A);
2) PC Checklist;
3) Trainer facility including:
    a. Underwater telephone log;
    b. Tactical book;
    c. Chronoscope;
    d. headsets
4) Sonar State Board;
5) Trainer Staff (min 1, preferred 2) and support personnel (2 if avail); and
6) NWO if avail to act as OOW.

IMPACT OF FAILURE: Remedial Training / Re-test

INVIGILATOR NOTES:

- Pause the run or set scenario between trainee assessments such that no pertinent contacts do not require immediate action
- Requests for NWO, Trainer Staff, and by-standers / extra support personnel should be submitted to Submarine Division Training Coordinator prior to course start.

1) PASS / FAIL CRITERIA: 6/7 SATs, the following drills and reactions must be achieved correctly by the SWL to be considered “satisfactory” on PC checklist contact type and voice report boxes (all 3 boxes):

- Proximity checks;
- Abrupt-in contact;
- Abrupt-in fishing vessel;
- Emergency go-deep;
- Torpedo;
- Aircraft on top;
- Transmission;
- Close quarter contact;
- Transients;

2) IMPACT OF FAILURE: Remedial training, re-test.

Sonar Watch Leader– EC 001.03 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

ITEM	PASS	FAIL	COMMENTS
Supervise Sound Room Drills and Procedures			
Monitor operation of submarine sonar equipment • Sonar System Management page • Array Health Charts			
Direct and supervise sound room drills and procedures as they arrive in scenario, such as: • New Contact • Proximity checks • Return to Periscope Depth Stern Arc clearance			
Maintain situational awareness of sonar activities (contact management)			
Provide Recommendations (Situation, Assessment, Recommendation)			
Assess contact type	☐	☐	
Provide voice report of contact Appropriate reporting procedure for contact type	☐	☐	
Voice report of contact follows correct format	☐	☐	
Recommendations • Logical • Supported by information • Relevant to scenario	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PERFORMANCE CHECK 001

PC Time: 600 mins (concurrent) + 50 min debrief

ASSESSMENT ACTIVITY: You are the Sonar Watch Leader (SWL) onboard a Victoria Class Submarine, the Submarine is operating in Canadian Territorial waters where you expect to detect merchant traffic and various threat platforms, you are about to assume a watch and fulfill your SWL duties.

Part 1: Turnover (10 minutes/trainee)
The trainee will begin watch turnover (oncoming) by conducting function check of sensors and equipment and then reporting for OOW brief.

Instructor will provide OOW brief (or OOW will be provided by NWO if available) to on-coming trainee

On-coming Trainee will review Sonar State Board and report for briefing from offcoming SWL (played by instructor if required or trainee off-going assessment)

Out-going trainee will complete watch turnover – offcoming

Part 2: SWL (90 minutes)
On-coming trainee will assume SWL and begin supervision of Sound Room drills and procedures.

2x By-standers (trainees not being assessed or external) will fill sonar operator positions and be directed by/supervised by SWL

Trainer staff (minimum 1, preferred 2) will operate trainer and progress scenario to include the following contact types as a minimum:
- Merchant vessel;
- Aircraft
- Warship
- Submarine
- Fishing vessel
- Pleasure craft
- Ferry
- Sonar buoy

Pass/fail criteria: 8/10 SATs, the following drills and reactions must be achieved correctly by the SWL to be considered “satisfactory” on PC checklist contact type and voice report boxes (all 3 boxes):
- Proximity checks;
- Abrupt-in contact;
- Abrupt-in fishing vessel;
- Emergency go-deep;
- Torpedo;
- Aircraft on top;
- Transmission;
- Close quarter contact;
- Transients;

ASSESSMENT RESOURCES:
- (Bulleted list of everything given to trainees for PC/EC)

IMPACT OF FAILURE: Remedial Training / Re-test

Sonar Watch Leader (AKYG)– PC 001 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES
In scenario: scenario will be stopped by instructor (assessment ended and failed) if SWL performance would put submarine safety in jeopardy or threat of harm to others, or if continued mal-performance constitutes negligence in opinion of instructor.

- At end of assessment: instructor will indicate to trainee whether they passed or failed and confirm that a detailed one-on-one feedback session will take place within 24 hours
- Detailed feedback with PC checklist will be provided one-on-one within 24 hours of assessment completion; (5 minutes per trainee, should happen during other scheduled activities)
- Group feedback / hot wash will occur at end of all assessments, where instructor will discuss what went well overall, overall points for improvement, and provide additional guidance as required
- Pause the run or set scenario between trainee assessments such that no pertinent contacts (do not require immediate action).
- Requests for NWO, Trainer Staff, and by-standers / extra support personnel should be submitted to Submarine Division Training Coordinator prior to course start.

3) PASS / FAIL CRITERIA: 8/10 SATs.

4) IMPACT OF FAILURE: IAW NTORD 4500-54.

ITEM	PASS	FAIL	COMMENTS
Conduct Watch Turnover IAW VCSSO AL6 - oncoming			
Function check of sensors and equipment • 2008 • Echo Sounder • 2004 • BQQ-10 Fire Control Display	☐	☐	
OOW Briefing, Sonar State Board review and brief from offcoming SWL	☐	☐	
Conduct Watch Turnover IAW VCSSO AL6 - offcoming			
Briefing oncoming SWL • Sensor status • Sonar picture	☐	☐	

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ITEM	PASS	FAIL	COMMENTS
• Other info as required Report turnover to OOW			
Supervise Sound Room drills and procedures			
Monitor operation of submarine sonar equipment	☐	☐	
Direct and supervise sound room drills and procedures	☐	☐	
Maintain situational awareness of sonar activities (contact management)	☐	☐	
Provide Recommendations (Situation, Assessment, Recommendation)			
Assess contact type	☐	☐	
Provide voice report of contact Appropriate reporting procedure for contact type	☐	☐	
Voice report follows correct format	☐	☐	
Recommendations • Logical • Supported by information • Relevant to scenario	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

Nil.

ANNEX K – TT PLAN

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta
Nil.			

MARPAC RDIMS #1402031
Chap 1 & 2 approved 13/12/2023
Chap 3 & 4 approved 18/01/2024

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

See QSWB Record of Decisions, Annex D.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	
Trg Mgr:	

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

A	
ASW	Anti-Submarine Warfare
ATT	Actual Training Time
B	
BAT	Base Administrative Time
BSC	Basic Submarine Course
C	
CAF	Canadian Armed Forces
CFITES	Canadian Forces Individual Training and Education System
Cmdt	Commandant
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
CSNI	Consolidated Secret Network Infrastructure
CSO	Captain’s Standing Orders
D	
DAOD	Defence Administrative Orders and Directives
DSCDS	Defence Subject Classification and Disposition System
DND	Department of National Defence
DPGR	Director Personnel Generation Requirements
DWAN	Defence Wide Area Network
E	
EC	Enabling Check
EDI	Explain, Demonstrate, Imitate
EO	Enabling Objective
G	
GPNet	General Purpose Network
H	
HOD	Head of Department
I	
IAW	In Accordance With
ID	Identification
IMS	Information Management Standard
IS	Information Systems
IT	Information Technology
ITEM	Interactive Electronic Technical Manual
L	
LO	Learning Objective
LG	Lesson Guidance

M	
MITE	Military Individual Training and Education
MOSID	Military Occupational Structure Identification Code
N	
NCPM	Naval Combat Operating Procedures Manual
NQual	National Qualification
NFS	Naval Fleet School
NAVORD	Naval Order
NTG	Naval Training Group
NPTORD	Naval Personnel and Training Order
NWO	Naval Warfare Officer
O	
OJT	On-the-Job Training
OJPR	On-the-Job Performance Requirement
OOW	Officer of the Watch
P	
PAL	Passive Aural Listening
PC	Performance Check
PO	Performance Objective
PLAR	Prior Learning Assessment and Recognition
PT	Physical Training
Q	
QSP	Qualification Standard and Plan
R	
R&Q	Rations & Quarters
RCN	Royal Canadian Navy
RSS	Request for Service Support
S	
SAR	Situation, Assessment, Recommendation
SAT	Satisfactory
SO	Standing Order
SOP SWL	Standard Operating Procedure Sonar Watch Leader
T	
TAT	Training Administrative Time
TA	Training Authority
TD	Temporary Duty
TE	Training Establishment
TEL	Technology Enabled Learning
TO	Training Officer
TPO	Training Petty Officer
TRB	Training Review Board
TS	Training Site
TTT	Training Travel Time
TWS	Tactical Weapons System

U	
UNCLAS	Unclassified
UNSAT	Unsatisfactory
UWW	Under Water Warfare
V	
VCSOP	Victoria Class Standard Operating Procedures
VCSSO	Victoria Class Ship’s Standing Orders
VCSCTT	Victoria Class Submarine Command Team Trainer
Vol	Volume
W	
WG	Working Group